Answer **one** question in this section in the spaces provided.

8 **(a)** The internal energy of an ideal gas is dependent on its state, and is given by the sum of the *random* kinetic energies of all its molecules.

(i) The state of an ideal gas depends on pressure, volume and two other quantities. Write down these two other quantities.

.....

..... [2]

(ii) Explain why it is important to include the word *random* in this definition.

.....

..... [1]

(iii) Explain why the potential energy of the molecules is not included in this definition.

.....

..... [1]

(iv) State two physical conditions under which a real gas will behave approximately as an ideal gas.

.....

..... [2]

(b) The pressure p exerted by an ideal gas is given by the equation

$$p = \frac{1}{3}\rho < c^2 >$$

(i) What do the symbols ρ and $<c^2>$ represent?

.....

..... [2]

(ii) Use this equation to derive an expression for the total internal energy of n moles of an ideal gas at temperature T.

[2]

(iii) Air contains oxygen and nitrogen molecules. Assuming air is an ideal gas, state and explain whether each of the following quantities is the same for oxygen and nitrogen molecules in air at a given temperature.

1. mean translational kinetic energy per molecule

.....

..... [1]

2. root mean square speed

.....

..... [1]

(c) A heat engine uses 10 moles of an ideal gas as a working substance. Fig. 8.1 shows the changes in pressure and volume of the gas during one cycle ABCA of operation of the engine.

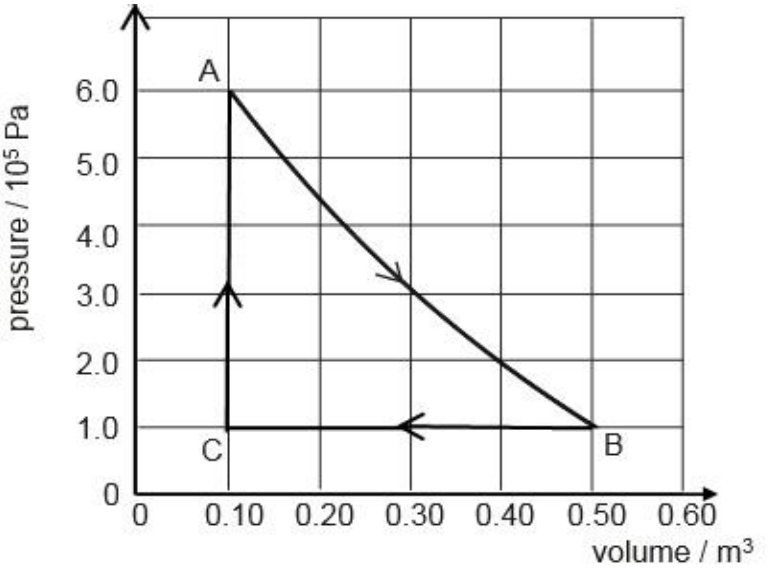

Fig. 8.1

(i) Using values from Fig. 8.1, calculate the temperature of the gas at point A.

temperature = ... K [2]

(ii) Show that the process A → B does not take place at a constant temperature.

.....

..... [1]

(iii) Use Fig. 8.1 to estimate the net work done by the gas during one cycle.

work done = ... J [2]

(iv) Hence, or otherwise, state the amount of heat absorbed by the gas during one cycle.

heat absorbed = ... J [1]

(v) If the temperature at A is maintained throughout the process A → B, state and explain how your answer in **(c)(iv)** may change.

.....

.....

.....

.....

..... [2]